

OXYGEN HOME



Why is this project a best practice?

Challenge

The project addresses the difficulty of scaling low-energy and healthy-building principles beyond a single dwelling. It had to combine density or phased delivery with reliable energy performance, good indoor conditions, landscape quality, mobility and construction cost, while keeping the sustainability measures practical for many future occupants.

Solution

The response combines heat pumps combined with low-temperature underfloor heating and cooling; mechanical ventilation with heat recovery and photovoltaic generation; and smart-home and building-automation controls intended to optimise comfort and energy use. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The approach can be scaled through repeatable specifications for envelope, MEP systems, materials, commissioning, mobility and landscape. Replication is strongest when the developer applies the same measurable requirements to every phase and verifies them at handover, rather than treating certification as a one-off design exercise.

Key facts and figures

Location	Voluntari / northern Bucharest, Romania
Building use / function	Premium single-family villa development
Project type (new build, refurbishment, extension)	New build; small gated community
Plot area / Floor area	12 villas reported; exact certified floor area not publicly disclosed
Year / phase	Green Homes-certified / phased development

Climate zone	Temperate continental
Thematic focus / innovation	Heat pumps; underfloor heating and cooling; heat-recovery ventilation; photovoltaics; smart-home controls
Investor / building owner	Private investor / Oxygen Home development
Architect	Not publicly disclosed in the consulted sources
Other involved stakeholders	Romania Green Building Council; HVAC, PV and building-automation specialists
Contact person	Oxygen Home - https://oxygen-home.ro/
Further information	https://green-homes.org/property/oxygen-home/ https://www.rogbc.org/proiecte-certificate-rogbc?lang=en https://oxygen-home.ro/

Oxygen Home is a new build; small gated community project in Voluntari / northern Bucharest, Romania. It functions as premium single-family villa development. Its main best-practice contribution is A luxury-villa project that makes integrated low-carbon building services central to the product offer. The documented strategy brings together heat pumps combined with low-temperature underfloor heating and cooling, mechanical ventilation with heat recovery and photovoltaic generation, and smart-home and building-automation controls intended to optimise comfort and energy use. Public marketing sources report 12 villas and a Green Homes approach; detailed audited performance was not located. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - Heat pumps combined with low-temperature underfloor heating and cooling.
2. Integrated systems and operation - Mechanical ventilation with heat recovery and photovoltaic generation.
3. Wider value - Smart-home and building-automation controls intended to optimise comfort and energy use.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.